

Cell Wall Dynamics and Biomolecular Condensates in Plants

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Abstract

Plant cell wall integrity and biogenesis depend on coordinated regulation of sensing and adaptive responses across developmental and stress contexts. The cell wall's dynamic architecture, composed of cellulose, hemicelluloses, pectins, and structural proteins, must continuously balance growth requirements with structural maintenance. This process encompasses biosynthetic control of wall polymers, mechanical sensing via plasma membrane receptors, signal transduction cascades, and (post-)transcriptional reprogramming of wall-modifying genes. Biomolecular condensates, membrane-less assemblies that organize cellular processes, have recently emerged as potential regulatory nodes in cell wall dynamics. Certain biomolecular condensates directly associate with wall synthesis, sensing, or repair mechanisms, while others contribute indirectly through osmotic regulation, mRNA sequestration, or transcriptional control of wall-modifying genes. Here we synthesize current evidence for condensate involvement in cell wall dynamics, distinguishing experimentally validated direct connections from indirect links and speculative hypotheses. We further discuss how artificial intelligence approaches could accelerate discovery of condensate-cell wall regulatory nodes and optimise plant performance under developmental and environmental constraints.

1. Introduction

The primary plant cell wall is a dynamic extracellular matrix that defines cell shape, mediates intercellular adhesion, and serves as the primary interface between plant cells and their environment (Cosgrove, 2015a). Composed mainly of cellulose microfibrils embedded in a matrix of hemicelluloses, pectins, and structural proteins, the cell wall provides mechanical support while remaining sufficiently plastic to accommodate cell expansion during growth (Du et al., 2020; Lin et al., 2022; Roeder et al., 2021; Wachsman et al., 2020). This composite architecture enables the wall to withstand turgor pressures that vary with cell type and developmental stage, typically ranging from 0.2 to 0.6 MPa in most cell types and potentially exceeding 1 MPa in specialized tissues (Beauzamy et al., 2014). The cell wall also functions as a selective physical barrier that influences the passage of water, nutrients, and signalling molecules, as well as a substrate for signal generation: modifications to wall composition or structure generated by mechanical stress, pathogen attack, or abiotic perturbations produce signals that are detected by plasma membrane-localized receptors, which then transduce these signals into intracellular cascades that integrate environmental information with developmental programmes (Franck et al., 2018). The composition and mechanical properties of cell wall are continuously remodelled through coordinated biosynthesis, secretion, assembly, and modification of its constituent polymers. These processes must be regulated to balance growth requirements with maintenance of structural integrity across diverse developmental contexts and environmental conditions (Hamann, 2015).

Biomolecular condensates (or simply ‘condensates’) are organelle-like assemblies that organize cellular biochemistry in all organisms, including plants (Kim et al., 2021; Solis-Miranda et al., 2023). Condensates comprise proteins usually rich in intrinsically disordered regions (IDRs) and typically form through the process of liquid-liquid phase separation (LLPS; for further discussion in the

phenomena that lead to condensate formation, we refer the reader to **Supplemental File 1** and the references therein) from proteins, RNAs, and metabolites that, upon specific stimuli, de-mix from the surrounding cytoplasm to create distinct compartments generally lacking membrane boundaries. Recent discoveries show that certain condensates require membrane surfaces for their function or even their formation (Jaillais and Ott, 2020; Snead and Gladfelter, 2019). One of condensates' most important features is their ability to rapidly assemble and disassemble. This reversibility is linked to environmental surveillance: specific signals can induce condensate formation and growth, or conversely, their dissolution. This responsiveness endows condensates with diverse signalling properties and a broad range of functions, from regulation of photosynthesis and metabolism to stress responses and developmental transitions (Gutierrez-Beltran et al., 2023; Huang and Xia, 2024).

Cell wall biosynthesis, trafficking, assembly, and remodelling require precise spatiotemporal coordination across multiple subcellular compartments. These processes must respond dynamically to developmental and environmental cues, posing a fundamental organisational challenge. Biomolecular condensates are increasingly recognised as regulatory hubs that help meet this challenge by concentrating enzymatic machinery, signalling components, and transcriptional regulators required for cell wall biogenesis and integrity maintenance. Consequently, condensates are particularly relevant for environmental surveillance by cooperating with, or directly participating in, the cell wall integrity (CWI) pathway. CWI collectively refers to a cell's ability to sense, maintain, and dynamically regulate cell wall structure and function in response to developmental cues or stress (Engelsdorf et al., 2018; Rui and Dinneny, 2020). Despite several recent discoveries implying connections between condensates and CWI maintenance, we lack guidelines and a clear conceptual framework for studying their relationship. We thus build on recent discoveries to suggest a research path forward for exploring the links between condensates and CWI. We also examine molecular mechanisms underlying condensate formation, distinguishing between condensates

directly and indirectly linked to CWI (Jung et al., 2020; Solis-Miranda *et al.*, 2023; Yuan and Zhao, 2025; Zhu et al., 2022). Finally, we analyse how proper usage of artificial intelligence (AI) approaches could accelerate discoveries in this field.

2. Principles of cell wall regulation and pathways underlying cell wall biogenesis

The wall undergoes continuous remodelling through cellulose synthesis, polysaccharide modification, and enzymatic loosening, enabling dynamic responses during growth and stress (Bou Daher et al., 2024; Cosgrove, 2016). Cell wall biogenesis requires the coordinated synthesis, trafficking, and assembly of distinct polysaccharide classes through spatially segregated biosynthetic pathways. Cellulose, the primary load-bearing component, is synthesized directly at the plasma membrane by cellulose synthase A (CESA) proteins assembled into hexameric rosette complexes (Gonneau et al., 2014; Paredez et al., 2006). Each rosette is proposed to contain approximately 18 CESA subunits (Hill et al., 2014), organized as a hexamer of trimers, extruding glucan chains, which immediately crystallise into microfibrils through hydrogen bonding and van der Waals interactions (Cosgrove, 2015b). The orientation of cellulose deposition is guided by cortical microtubules, which constrain rosette movement and thereby determine the directionality of cell expansion (Paredez et al., 2008; Paredez *et al.*, 2006). CESA complexes are delivered to the plasma membrane via Golgi-derived vesicles and internalized through clathrin-mediated endocytosis, with continuous cycling between the plasma membrane and *trans*-Golgi network/early endosome compartments (TGN/EE) maintaining a dynamic pool of active synthase complexes (Bashline et al., 2014).

In contrast to cellulose, matrix polysaccharides including hemicelluloses and pectins are directly synthesized within Golgi cisternae by membrane-bound glycosyltransferases (Bou Daher *et al.*, 2024; Rajaraman *et al.*, 2016; Tatenno *et al.*, 2015; Yang *et al.*, 2016). Hemicellulose synthesis involves CELLULOSE SYNTHASE-LIKE (CSL) family proteins that produce diverse polysaccharides including xyloglucans, mannans, and mixed-linkage glucans, with specific CSL clades (e.g., CSLA mannans, CSLF/CSLG mixed glucans) with distinct substrate specificities and developmental expression patterns. Pectin biosynthesis proceeds through sequential glycosyltransferase activities that elaborate homogalacturonan, rhamnogalacturonan I, and rhamnogalacturonan II backbones with various side-chain modifications (Haas *et al.*, 2021). These matrix polymers are packaged into secretory vesicles that traffic along actin filaments to fusion sites at the plasma membrane, where vesicle tethering complexes and SOLUBLE N-ETHYLMALEIMIDE-SENSITIVE FACTOR ATTACHMENT PROTEIN RECEPTOR (SNARE) proteins mediate targeted exocytosis (Bashline *et al.*, 2014; Sinclair *et al.*, 2018).

The assembly of delivered components into functional wall architecture requires coordinated polymer interactions and enzymatic modifications at the cell surface. Hemicelluloses such as xyloglucan bind to cellulose microfibrils through hydrogen bonding, cross-linking adjacent microfibrils and modulating wall extensibility (Rosquete and Drakakaki, 2018; Sinclair *et al.*, 2018). Pectins form gel-like networks through Ca^{2+} -mediated cross-linking of demethylesterified homogalacturonan domains and borate ester bridges in rhamnogalacturonan II, establishing a matrix that embeds cellulose-hemicellulose networks. Wall-localized enzymes including xyloglucan endotransglucosylase/hydrolases (XTHs), pectin methylesterases (PMEs), and expansins continuously remodel these networks by cleaving and reforming glycosidic linkages, modifying pectin methylesterification patterns, and disrupting hydrogen bonds, thereby enabling wall loosening during cell expansion (Cosgrove, 2015a). The spatial distribution of these remodelling activities

creates localised zones of differential wall extensibility that direct anisotropic growth and morphogenesis (Peaucelle et al., 2011; Uyttewaal et al., 2012).

Beyond primary wall synthesis during growth, plant development requires specialized biosynthetic programs that generate functionally distinct wall architectures adapted to specific cell types and developmental contexts. Secondary cell walls, deposited between the plasma membrane and primary wall after cell expansion ceases, provide enhanced mechanical strength and water impermeability essential for vascular transport and structural support (Ali et al., 2025). Secondary wall formation involves distinct CESA isoforms (CESA4, CESA7, and CESA8 in *Arabidopsis thaliana*) that assemble into rosette complexes producing cellulose microfibrils with higher crystallinity and altered orientation compared to primary walls (Hamann, 2012; Nakano et al., 2015; Zhu et al., 2025). The secondary wall matrix is enriched in xylans synthesized by IRREGULAR XYLEM (IRX) family glycosyltransferases and, in fibre cells and vessels, extensively lignified through oxidative polymerisation of monolignol precursors exported by ATP-binding cassette (ABC) transporters (Kaneda et al., 2011; Prats Mateu et al., 2016; Van der Does et al., 2017b; Voiniciuc et al., 2013). Lignin deposition proceeds through laccase and peroxidase-catalysed radical coupling reactions that generate a hydrophobic, mechanically rigid polymer interpenetrating the polysaccharide network (Jacobowitz et al., 2019; Lee et al., 2018; Peralta Ogorek et al., 2025; Zhu et al., 2025). This secondary wall biosynthetic program is orchestrated by a hierarchical transcriptional network. NO APICAL MERISTEM (NAC) master regulators, like the SECONDARY WALL-ASSOCIATED NAC DOMAIN PROTEIN 1 (SND1) and VASCULAR-RELATED NAC-DOMAIN (VND) proteins, initiate the cascade by activating downstream MYELOBLASTOSIS (MYB) transcription factors that directly induce cellulose synthase, xylan synthase, and lignin biosynthetic genes (Nakano et al., 2015; Zhong et al., 2007b).

117 Apart from the secondary wall, pollen grains exemplify extreme specialisation in wall architecture,
118 with a composite structure comprising an inner intine layer and an outer exine layer serving distinct
119 functional requirements for desiccation tolerance, pollen tube emergence, and species-specific
120 recognition (Zhu et al., 2017). The intine, laid down by the microspore itself, resembles primary walls
121 in containing cellulose, hemicelluloses, and pectins but exhibits modified composition including
122 enhanced callose deposition and localized pectin modifications that create zones of differential
123 rigidity essential for controlled pollen tube emergence (Xu et al., 2024b). The exine, derived from
124 tapetal cell secretions, consists of sporopollenin, an extraordinarily resistant biopolymer of
125 hydroxylated and carboxylated aliphatic chains cross-linked with phenolic moieties. Exine
126 patterning is determined by the primexine, a transient callose wall that templates subsequent
127 sporopollenin deposition through poorly understood mechanisms involving lipid transfer proteins,
128 (ATP-BINDING CASSETTE) ABC transporters, and cytochrome P450 enzymes that deliver and
129 polymerise sporopollenin precursors at specific deposition sites (Haas *et al.*, 2021; Zhu et al., 2020).
130 This remarkable patterning generates species-specific surface architectures with baculae,
131 columellae, and tectum structures sculpted at submicron resolution, enabling pollen recognition and
132 adhesion to compatible stigmatic surfaces (Jacobowitz *et al.*, 2019).

133 Another specialised cell wall-type is the seed coat. Seed coat cells produce highly specialised
134 mucilage layers comprising modified pectins that undergo explosive hydration upon imbibition,
135 generating a hydrated gel that facilitates seed dispersal and germination (Yang et al., 2019).
136 Mucilage is synthesized during seed maturation through upregulated expression of
137 rhamnogalacturonan I biosynthetic machinery enriched in MUCILAGE-MODIFIED (MUCI) and
138 MUCILAGE-RELATED (MUCR) glycosyltransferases that elaborate extensively branched
139 rhamnogalacturonan I with arabinose and galactose side chains (Voiniciuc et al., 2015). This
140 mucilage is packaged within columella, specialized cytoplasmic domains that organize mucilage into
141 radial layers beneath the outer tangential wall and collapse upon imbibition to release the hydrated

polysaccharide (Haughn and Western, 2012). Concurrent with mucilage synthesis, seed coat epidermal cells deposit secondary walls enriched in cellulose and lignin–like phenolics that reinforce the seed coat and resist microbial degradation, with the transcriptional regulator TRANSPARENT TESTA GLABRA 2 (TTG2) coordinating both mucilage biosynthesis and secondary wall formation programs (Li et al., 2019). The outer primary wall also undergoes extensive pectin modification through pectin methylesterases and pectin methylesterase inhibitors, creating demethylesterified pectin domains that serve as attachment sites for the released mucilage polysaccharides (Hongo et al., 2012).

3. Principles of cell wall integrity during stress

We structure our discussion of plant CWI around four functional components: (1) Perturbations (mechanical, osmotic, or chemical stresses affecting the wall); (2) Primary sensors (plasma membrane–localized receptor–like kinases and ion channels); (3) Signal transduction pathways (calcium fluxes, ROS production, cytoskeletal dynamics); and (4) Cellular responses (transcriptional, enzymatic, and developmental). We also include processes that blur the traditional distinction between development and stress, including pollen tube growth (developmental but involves mechanical stress), fruit ripening (developmental but involves ethylene stress), root hair formation (developmental but involves wall remodelling), and secondary wall deposition (developmental but involves massive restructuring).

While the most well–studied CWI, that of yeast centres on the RAS HOMOLOG 1–PROTEIN KINASE C 1–MITOGEN–ACTIVATED PROTEIN KINASE (Rho1–PKC1–MAPK) signalling module (Levin, 2011), plant CWI employs a more distributed network of receptor–like kinases and

interconnected signalling pathways. Multiple types of perturbations trigger CWI responses. Cellulose deficiency, caused by inhibition of cellulose synthase activity or genetic mutations, triggers compensatory responses including growth inhibition and ectopic lignification (Bou Daher *et al.*, 2024; Gonneau *et al.*, 2018b). Pectin modification states, particularly the degree of methylesterification, alter wall mechanical properties and signalling potential (Lin *et al.*, 2022; Moussu *et al.*, 2023). Physical damage, turgor pressure changes, and externally applied force disrupt the wall–membrane continuum, generating mechanical signals that activate stretch–sensitive channels and mechanosensitive receptors (Yoshimura *et al.*, 2021). Mechanical damage or enzymatic degradation releases damage–associated molecular patterns (DAMPs) including oligogalacturonides (OGAs) from pectin degradation, cellobioses (cellobiose, cellobiose) from cellulose breakdown, and xylo–oligosaccharides from hemicellulose hydrolysis (Brutus *et al.*, 2010; Tseng *et al.*, 2022). These DAMPs serve as endogenous danger signals. The WALL–ASSOCIATED KINASES WAK1 and WAK2 bind oligogalacturonides released during mechanical damage, activating CWI responses that include transcriptional induction of cell wall–related genes (Brutus *et al.*, 2010; Kohorn and Kohorn, 2012). Fungal and bacterial pathogens produce cell wall–degrading enzymes that release DAMPs and directly compromise wall integrity, triggering defence–related CWI responses (Ngou *et al.*, 2022; Savatin *et al.*, 2011; Savatin *et al.*, 2014).

Plant cells use multiple classes of receptor–like kinases (RLKs) to monitor CWI. FERONIA (FER) acts as a primary regulator of cell wall–plasma membrane signalling, sensing cell wall pectin status and transducing mechanical signals into intracellular responses (Franck *et al.*, 2018; Tang *et al.*, 2022). FER contains two tandem malectin–like ectodomains that bind demethylesterified pectin, directly monitoring wall properties via pectin methylesterification status (Lin *et al.*, 2022)(**Fig. 1**). FER participates in RALF–pectin condensates at the cell wall–plasma membrane interface, where these extracellular assemblies concentrate signalling components and modulate receptor activation (Liu *et al.*, 2024b; Zhang *et al.*, 2024). THESEUS1 (THE1), another CrRLK1L family member with

malectin domains, senses cellulose deficiency (**Fig. 1**). THE1 activates compensatory responses including ectopic lignification and growth inhibition when cellulose synthesis is compromised (Bacete et al., 2022; Gigli-Bisceglia et al., 2022; Gonneau et al., 2018b; Hematy et al., 2007). THE1 functions as a receptor for RAPID ALKALINIZATION FACTOR 34 (RALF34) peptide (Gonneau et al., 2018a) and responds to turgor pressure changes, such as those occurring during osmotic stress, thereby linking osmotic pressure changes to cell wall sensing (Bou Daher et al., 2024).

WAK1 and WAK2 except oligogalacturonides binding, they provide additional CWI sensing through binding pectin via both covalent and Ca^{2+} –dependent ionic interactions (Brutus et al., 2010; Kohorn and Kohorn, 2012)(**Fig. 1**). WAKs bind oligogalacturonides released during mechanical damage or pathogen attack, activating CWI responses (Kohorn and Kohorn, 2012; Liu et al., 2023a). The dual binding mechanism, covalent attachment to intact pectin and reversible ionic binding to pectin fragments, enables WAKs to distinguish between normal wall status and damage states.

CORK1 (also known as IGP1), IGP3, and IGP4 constitute a subfamily of LRR–malectin receptor kinases that detect celooligomers and xylo–oligosaccharides released during wall degradation (Martín-Dacal et al., 2023; Tseng et al., 2022). These receptors provide specific recognition of hemicellulose and cellulose breakdown products, complementing the pectin–sensing capacity of WAKs.

MCA proteins (Mid1–Complementing Activity) at the plasma membrane function as mechanosensitive Ca^{2+} channels that respond to membrane tension changes (Yoshimura et al., 2021). These single–transmembrane proteins form homotetramers that facilitate calcium influx upon mechanical perturbation (**Fig. 1**). This arrangement provides a direct mechanism for converting physical force into biochemical signals, positioning MCAs as primary mechanotransducers at the plasma membrane.

The CWI receptor systems described above function cooperatively, with co-receptors like BRI1-ASSOCIATED RECEPTOR KINASE 1 (BAK1) and downstream kinases like BOTRYTIS-INDUCED KINASE 1 (BIK1) integrating signals from multiple RLKs to coordinate CWI maintenance with growth, development, and stress responses. BAK1, for example, integrates CWI with immunity (Engelsdorf *et al.*, 2018). Additional receptor systems contribute to CWI sensing. The FERONIA RELATED LRR-RLKs FEI1/FEI2 function downstream of THE1 and upstream of MCA1, sensing the arabinogalactan protein FASCICLIN-LIKE ARABINOGALACTAN PROTEIN 4 / SALT OVERLY SENSITIVE 5 (FLA4/SOS5, **Fig. 1**) to regulate cellulose biosynthesis via modulating ethylene signalling through the AMINOCYCLOPROPANE-1-CARBOXYLIC ACID SYNTHASE (ACC synthase) (Engelsdorf *et al.*, 2018; Xu *et al.*, 2008). MALE DISCOVERING KINASE 2 / LEUCINE-RICH REPEAT-KINASE INVOLVED IN STRESS SIGNALING (MIK2/LRR-KISS) serves as a critical nexus linking CWI sensing to immunity by perceiving SERINE-RICH ENDOGENOUS PEPTIDE (SCOOP) peptides, particularly SCOOP18, which is transcriptionally induced upon wall damage (Van der Does *et al.*, 2017a).

Signal transduction from CWI sensors involves multiple interconnected pathways that rapidly convert wall perturbations into cellular responses. Multiple additional channels, apart from the above mentioned MCAs, mediate Ca^{2+} influx in response to CWI perturbations. The above-mentioned MCA channels respond directly to membrane tension, providing rapid Ca^{2+} influx during mechanical stress (Yoshimura *et al.*, 2021). Cyclic nucleotide-gated channels (CNGC2/4/20 heteromers) form heterotetrameric calcium-permeable channels regulated by calmodulin and activated by BIK1 kinase phosphorylation (Ma and Berkowitz, 2011; Tian *et al.*, 2019). These channels integrate pathogen-associated molecular pattern (PAMP) signalling with CWI responses. Hyperosmolality-gated Ca^{2+} -permeable channels (OSCA) family members mediate osmosensitive Ca^{2+} influx in response to osmotic stress (Yuan *et al.*, 2014), with OSCA1.3 playing a specific role in stomatal immunity (Thor *et al.*, 2020). OSCA channels link osmotic pressure, to Ca^{2+} -dependent signalling

cascades. The Ca^{2+} signals generated by these channels activate downstream calcium-binding proteins including Ca^{2+} -dependent protein kinases (CPKs) and calcineurin B-like (CBL) proteins that interact with CBL-interacting protein kinases (CIPKs), propagating signals to diverse cellular targets (Kadota et al., 2014).

NADPH oxidase RBOHD (Respiratory Burst Oxidase Homolog D) generates spatially restricted reactive oxygen species (ROS) bursts at sites of CWI perturbation (**Fig. 1**). RBOHD activity is regulated through multiple mechanisms: Ca^{2+} binding to N-terminal EF-hand motifs, and phosphorylation by BIK1 and CPK5 (Dubiella et al., 2013). Related NADPH oxidases RBOHF and RBOHC are additionally regulated by CBL1/9-CIPK26 complexes (Drerup et al., 2013; Zhang et al., 2018), suggesting similar regulatory mechanisms may operate for RBOHD. RBOHD-mediated ROS production creates signalling waves that propagate from damage sites at velocities up to 8.4 cm/min, enabling rapid systemic responses to local wall damage (Miller et al., 2009). These ROS waves coordinate cellular responses across tissues and provide oxidative signals that can trigger protein oxidation, modify transcription factor activity, and may influence condensate formation.

Although the specific MAPK modules activated during plant CWI responses remain incompletely characterized, evidence suggests involvement of MPK3/MPK6 cascades downstream of DAMP perception and ROS production (Ngou *et al.*, 2022). These kinases phosphorylate transcription factors and other regulatory proteins to implement adaptive responses.

The cytoskeleton serves dual functions in CWI: as a structural organizer of cell wall synthesis and as a dynamic relay system that responds to and propagates CWI signals. As mentioned above, cortical microtubules guide cellulose synthase complex (CSC) trajectories in the plasma membrane ('alignment hypothesis'), thereby controlling the orientation of nascent cellulose microfibrils (Paredes *et al.*, 2006). This microtubule-cellulose alignment is fundamental to directed cell growth and wall anisotropy. The CELLULOSE SYNTHASE INTERACTING1 (CSI1) protein, also known as POM-

POM2, mediates the physical linkage between CSCs and cortical microtubules (Bringmann et al., 2012; Li et al., 2012). CSI1 contains Armadillo repeats typical of microtubule-binding proteins and co-localizes with both primary wall CSCs and cortical microtubules. CSI1 travels bidirectionally along microtubules at velocities of 200–350 nm/min (based on an average motility from the two previous studies), matching CSC speeds. In *csi1* mutants, CSC trajectories become uncoupled from underlying microtubules, and cellulose microfibrils adopt default transverse orientations regardless of microtubule organisation. Furthermore, genetic evidence demonstrates bidirectional communication between cellulose synthesis and microtubule organization. Mutations in PROCUSTE1 and KORRIGAN, both encoding proteins involved in cellulose synthesis, cause altered cortical microtubule orientation (Paredez et al., 2008).

Beyond trajectory guidance, cortical microtubules position delivery of CSCs to the plasma membrane and interact with CESA-containing trafficking compartments through mechanisms involving microtubule depolymerisation-driven motility (Gutierrez et al., 2009). During osmotic stress and other conditions that limit cellulose synthesis, association of CESA compartments with cortical microtubules increases dramatically, suggesting microtubules sequester CSCs when wall synthesis is inhibited. Cortical microtubules influence not only microfibril orientation but also cellulose crystallinity. The abundance of cortical microtubules enables decreases in wall crystallinity during rapid growth at high temperatures (Fujita et al., 2011; Fujita et al., 2013). Reduced microtubule mass in the temperature sensitive *Arabidopsis* mutant allele *microtubule organization 1 (mor1-1)* at high temperatures prevents normal crystallinity decreases and causes loss of unidirectional expansion, demonstrating that microtubules control physico-chemical properties of cellulose essential for wall integrity.

The plant-specific ROP (Rho-of-Plants) GTPases act as molecular switches that regulate actin organisation, membrane trafficking, ROS production, and growth polarity in response to

developmental and environmental signals (Payne and Grierson, 2009; Yang et al., 2007; Zhang et al., 2013). CWI sensors including FERONIA directly activate ROPs through physical interaction with ROP guanine nucleotide exchange factors (ROPGEFs), establishing ROPs as central integrators that transduce cell wall pectin status and mechanical signals into cytoskeletal reorganization (Duan et al., 2010; Lin et al., 2022). The EDICATOR OF CYTOKINESIS (DOCK) family guanine nucleotide exchange factor SPIKE1 activates ROPs, which then associate with the WAVE/SCAR (WASP-FAMILY VERPROLIN-HOMOLOGOUS PROTEIN/ SUPPRESSOR OF CYCLIC AMP RECEPTOR) regulatory complex. Activated WAVE/SCAR stimulates the ACTIN-RELATED PROTEIN 2/3 (ARP2/3) complex to nucleate branched actin filaments (Zhang et al., 2013). This pathway is essential for cell morphogenesis, with mutations causing distorted trichomes and pavement cells. ROPs regulate phosphorylation of ACTIN-DEPOLYMERIZING FACTORS (ADFs), controlling their actin-severing and depolymerizing activities (Chen et al., 2003; Hussey et al., 2002; Nan et al., 2017; Qian et al., 2023). Salt and osmotic stresses, which affect CWI by altering turgor pressure and membrane-wall interactions, induce rapid actin reorientation mediated by ADFs (Wang et al., 2021a). Furthermore, Ca^{2+} influx through MCA, CNGC, and OSCA channels activates the above-mentioned Ca^{2+} -binding proteins that modulate actin-binding protein activity. The mechanistic connections include: (1) direct Ca^{2+} binding to actin-binding proteins, which alters their activity; (2) CALCIUM-DEPENDENT PROTEIN KINASE (CDPK)-mediated phosphorylation of ADFs and other actin regulators; (3) ROP GTPase activation in response to CWI signals, which subsequently modulates actin nucleation and turnover; and (4) CBL/CIPK pathway regulation of actin organization during salt stress (Ye et al., 2013).

The actin cytoskeleton provides structural organization essential for CWI maintenance while simultaneously serving as a dynamic relay system that coordinates with condensate-forming components. ROP (Rho-of-Plants) GTPases function as central integrators connecting cell wall status to cytoskeletal dynamics. ROP6 exemplifies this integration by simultaneously regulating

actin organization and microtubule severing through the ROP6–ROP–INTERACTIVE CRIB MOTIF–CONTAINING PROTEIN 1–KATANIN pathway (Lin et al., 2013; Poraty-Gavra et al., 2013). The functional importance of this coordination is evident in dominant–negative *rop6* mutants, which show altered actin and microtubule organisation accompanied by twisted leaves and constitutive defence responses, phenotypes indicating disrupted CWI maintenance. This pleiotropic phenotype suggests that ROPs serve as critical nodes integrating mechanical signals from the cell wall with cytoskeletal organisation and stress signalling. The molecular pathways linking CWI receptors to ROP activation involve multiple second messengers that also regulate condensate behaviour, creating potential coordination between cytoskeletal dynamics and phase separation events. While direct biochemical connections between specific CWI receptors (FER, THE1, WAKs) and ROPGEFs require further characterization, the pathway architecture suggests a multi–component signalling cascade.

The integrated signalling pathways described above culminate in coordinated transcriptional and post–transcriptional responses that adjust cell wall properties and cellular physiology. However, these responses must be carefully partitioned into those directly sensing cell wall status versus those responding to secondary consequences of CWI perturbations, particularly osmotic stress. Direct CWI transcriptional responses involve transcription factors activated by cell wall damage or modification. MYB transcription factors, particularly MYB46 and MYB83, function as master regulators of secondary cell wall biosynthesis, coordinating expression of cellulose synthases, xylan biosynthetic enzymes, and lignin biosynthetic genes (Zhong et al., 2007a). These transcription factors activate compensatory lignification when primary wall synthesis is compromised, providing alternative mechanical support, a response directly targeting CWI defects (Gonneau *et al.*, 2018a). Similarly, THE1 receptor activation by cellulose deficiency triggers growth inhibition to prevent cell expansion when walls cannot provide adequate mechanical support, representing a direct CWI–protective response (Hematy *et al.*, 2007).

NAC family transcription factors, including members of the NO APICAL MERISTEM (NAM), ARABIDOPSIS TRANSCRIPTION ACTIVATION FACTOR (ATAF), and CUP-SHAPED COTYLEDON (CUC) subfamilies, regulate various stress responses and secondary metabolism (Nakano *et al.*, 2015), though their specific roles in CWI-related gene expression require further characterization. When CWI perturbations result from pathogen attack, defence-specific transcriptional programs activate, inducing callose deposition, antimicrobial compound synthesis, and systemic acquired resistance (Ngou *et al.*, 2022).

Enzymatic responses downstream of transcriptional activation enable dynamic cell wall remodelling. Expansins loosen wall structure by disrupting non-covalent bonds between cellulose microfibrils and matrix polysaccharides, facilitating cell expansion during growth (Cosgrove, 2016). PMEs modulate the degree of pectin methylesterification, altering wall mechanical properties, Ca^{2+} cross-linking capacity, and the availability of demethylesterified pectin for interaction with CWI receptors like FER (Hocq *et al.*, 2016; Sénéchal *et al.*, 2014). XTHs cut and re-ligate xyloglucan chains, enabling controlled wall restructuring during both normal growth and stress responses (Sharples *et al.*, 2017). The coordinated regulation of these enzymes allows cells to fine-tune wall properties in response to developmental signals and environmental challenges while maintaining the mechanical integrity essential for survival.

4. Condensates in cell wall regulation and pathways underlying cell wall biogenesis

The coordination of cell wall biogenesis across multiple subcellular compartments, requires spatial organisation of biosynthetic machinery, trafficking components, and regulatory factors. Condensates provide a potential mechanism to concentrate functionally related proteins and

achieve such spatiotemporal coordination. In mammalian secretory systems, condensates organise diverse processes including Golgi cisterna biogenesis, cargo sorting at the TGN, and receptor clustering at the plasma membrane. Whether analogous condensate-based mechanisms organise plant cell wall biosynthetic pathways remains largely unexplored, though emerging evidence suggests phase separation may operate at multiple levels from polysaccharide synthesis through cytoskeletal guidance of wall deposition. Below, we examine how condensates contribute to these pathways, without distinguishing between condensate's types and how they have formed. For more information on condensates' dynamics, we provide a thorough analysis of condensate's behaviours and formation pathways as **Supplemental File 1**.

4.1 Extracellular condensates in cell wall biogenesis

Plant cell walls are not passive structural matrices but rather dynamic assemblies where polysaccharides and associated proteins can undergo phase separation to create functionally specialized domains. Unlike cytoplasmic condensates composed primarily of proteins and nucleic acids, these extracellular assemblies involve heterotypic interactions between polysaccharides, particularly pectins, and peptide ligands, representing an atypical form of biomolecular condensation with distinct material properties. The most comprehensively characterized extracellular condensate system involves interactions between RALF peptides and demethylesterified pectin. RALF peptides have IDRs and are positively charged with the ability to bind demethylesterified (negatively charged) homogalacturonan in the cell wall through direct electrostatic interactions (Liu et al., 2024c; Moussu et al., 2023). This pectin-RALF interaction undergoes LLPS in the extracellular space, forming condensates that recruit the plasma membrane receptor kinase FERONIA (FER) and its co-receptor LLG1 (LORELEI-LIKE GPI-ANCHORED PROTEIN 1) into pectin-RALF-FER-LLG1 assemblies at the cell wall-plasma membrane interface (Liu et al., 2024c). These condensates exhibit characteristic properties of phase-separated systems including concentration-dependent assembly, dynamic protein recruitment, and responsiveness to environmental conditions. This

finding establishes RALF–pectin condensates as a paradigm for how extracellular phase separation can organise receptor–mediated signalling at the cell surface, although whether other plant receptor systems employ similar condensate–based mechanisms remain to be determined.

In pollen tubes, the LRX8 protein (LEUCINE–RICH REPEAT EXTENSIN 8) forms a complex with RALF4 peptide that interacts with demethylesterified pectin in the cell wall (Moussu *et al.*, 2023). The LRX8–RALF4 complex adopts a heterotetrametric configuration in which two LRX8 molecules each bind one RALF4 peptide, and this tetrameric complex binds to demethylesterified pectin (i.e., negatively charged produced by PME) through charge–dependent interactions mediated by RALF4’s polycationic surface. This LRX8–RALF4–pectin interaction exerts a ‘condensing effect’ on the cell wall, organizing pectin polymers into a reticulated network with a dendritic distribution pattern that is essential for maintaining CWI and controlling pollen tube growth velocity (Moussu *et al.*, 2023). Pollen tubes expressing RALF4 mutants lacking the positively charged patch required for pectin binding display unstructured cell wall organization, accelerated growth velocity, and premature tube bursting, demonstrating that the spatial patterning created by LRX–RALF–pectin condensates is functionally critical. Similarly, in root hairs, the LRX1 and LRX2 proteins form complexes with RALF22 peptide that organize pectin into circumferential rings at the root hair tip (Schoenaers *et al.*, 2024). These LRX–RALF22–pectin condensates create periodic patterns that regulate polar growth and maintain cell wall assembly during tip expansion.

The material properties of these extracellular condensates differ from typical cytoplasmic phase–separated assemblies in several respects. First, they involve heterotypic interactions between proteins (RALF peptides, LRX proteins) and polysaccharides (demethylesterified pectin), creating hybrid assemblies where the polysaccharide network provides a charged scaffold that concentrates and organises peptide ligands. Second, their formation depends critically on the methylesterification status of pectin, which is dynamically regulated by PMEs that remove methyl groups from

homogalacturonan and PME inhibitors (PMEIs) that modulate PME activity (Daher and Braybrook, 2015; Rößling et al., 2024). This provides a regulatory mechanism where enzymatic modification of the polysaccharide scaffold controls condensate assembly and disassembly. Third, these condensates operate at the cell wall–plasma membrane interface, creating a physical platform that bridges extracellular matrix organization with intracellular signalling. The recruitment of membrane–localised receptors (FER, LLG1) into extracellular pectin–RALF condensates enables the cell to couple cell wall status directly to receptor activation and downstream signalling cascades. The methylesterification state of pectins is determined by the balance between PMEs and PME inhibitors (PMEIs) that form stoichiometric complexes with PMEs (Daher and Braybrook, 2015). Whether PME–PMEI complexes further organise through phase separation requires investigation, though PME–PMEI complex formation can function through conventional mechanisms.

The charge–dependent nature of pectin–RALF interactions provides spatial specificity through localised pectin demethylesterification. In pollen tubes, PME activity generates demethylesterified pectin domains specifically in the subapical region where cell wall reinforcement is required, creating a spatial template for LRX–RALF condensate assembly (Moussu *et al.*, 2023). The application of exogenous demethylesterified oligogalacturonides can compete with cell wall pectin for RALF binding, demonstrating that the interaction depends on the availability of negatively charged pectin (Rößling *et al.*, 2024).

4.2 Condensates and phase separation organising cell wall biosynthetic machinery

Polysaccharide biosynthesis in the Golgi requires sequential activities of glycosyltransferases that elaborate specific wall polymers through processive addition of sugar residues to growing chains. The spatial organization of these biosynthetic enzymes within Golgi cisternae remains incompletely understood but could in theory involve condensate–based mechanisms that concentrate functionally related glycosyltransferases. In mammalian cells, Golgi–resident glycosylation enzymes are

organised into discrete ‘zones’ within Golgi units that exhibit dynamic behaviour and differential mobility, and enzyme compartmentalisation across Golgi cisternae enables controlled assembly of glycan structures (Harada et al., 2024; Jaiman and Thattai, 2020). However, whether these organisational features involve phase separation remains to be determined. While condensates could provide a mechanism to concentrate functionally related proteins in biosynthetic organelles, direct evidence in plants remains limited to RHM1 ‘rhamnosome’ formation (Field et al., 2024), preprint). This work demonstrates that RHM1 forms enzymatically active rhamnosomes essential for UDP–rhamnose synthesis and pectin biosynthesis, providing the first direct evidence of LLPS organising plant cell wall metabolism. Whether similar condensate–based organization extends to other aspects of polysaccharide biosynthesis awaits investigation.

Cellulose synthase–like proteins, particularly CELLULOSE SYNTHASE–LIKE FAMILY F6 (CSLF6) involved in mixed–linkage glucan synthesis, localise to the Golgi apparatus where they function in MIXED–LINKAGE GLUCAN (MLG) biosynthesis (Kim et al., 2018). Whether these CSLF6 puncta represent phase–separated compartments requires characterisation using condensate criteria such as fusion behaviour, internal dynamics measured by Fluorescence recovery after photobleaching (FRAP), and sensitivity to hexanediol. Several pectin biosynthetic enzymes contain predicted IDRs (e.g., GAE1, QUASIMODO3; **Supplemental File 2**) that could theoretically mediate multivalent interactions, though IDR presence alone does not demonstrate phase separation capability.

Following biosynthesis, the TGN functions as a critical sorting hub where newly synthesized cell wall polysaccharides and modifying enzymes are packaged into distinct vesicle populations destined for the plasma membrane (Bashline et al., 2014). In mammalian cells, cargo sorting at the TGN involves condensates formed by proteins including chromogranins, secretogranins, and TGN46, which concentrate secretory cargo through multivalent protein–lipid interactions (Lujan et al., 2024; Parchure et al., 2022). Whether analogous mechanisms operate in plant TGN sorting of cell wall

polysaccharides and CESA complexes remains untested. Plant TGN compartments sort diverse cargo including cellulose synthase complexes, pectin-modifying enzymes, and cell wall polysaccharides into vesicles targeted to specific plasma membrane domains, requiring RAS-RELATED PROTEIN (RAB) GTPases (particularly RABA family members), SNARE proteins (SYP4 family syntaxins, Vesicle-Associated Membrane Protein VAMP72 v-SNAREs), and multi-subunit tethering complexes including Transport Protein Particle II (TRAPP II) and the exocyst (Bashline et al., 2015; Garcia et al., 2020; Jaber et al., 2010). Several of these components form punctate accumulations at TGN membranes that may represent phase-separated domains, though further validation with the criteria mentioned above is pending.

4.3 Membrane organization of condensates involved in cell wall biosynthesis

CESA complexes are not uniformly distributed but instead exhibit clustered distributions at the plasma membrane, with individual rosettes often appearing in proximity to one another (Gu et al., 2010). Whether this clustering reflects passive crowding within membrane domains or depends on organization through phase separation mechanisms (either of CESA or other 'helper' molecules) remains unclear. Recent work identified CSC nanodomains as key mediators of cell wall-plasma membrane attachments that confer hyperosmotic stress resilience (Rui et al., 2025), preprint). These CSC nanodomains, dynamically pattern wall-membrane contacts, suggesting a potential condensate-like role in stress-responsive cellulose organization, though LLPS remains untested. Furthermore, CSI1 forms discrete puncta at sites of active cellulose synthesis (Li et al., 2012; Liu et al., 2023d; Zhu et al., 2018). These CSI1 puncta could potentially represent phase-separated assemblies that coordinate CESA-microtubule coupling, though alternative explanations including conventional protein complex assembly are equally plausible. CSI1 contains predicted intrinsically disordered regions that could theoretically mediate multivalent interactions enabling phase separation.

4.4 Cytoskeletal condensates organising cell wall biosynthesis

Cortical microtubules guide cellulose synthase complex trajectories at the plasma membrane, thereby determining cellulose microfibril orientation and the mechanical anisotropy of cell walls that directs anisotropic cell expansion (Paredes *et al.*, 2006). The organization of cortical microtubule arrays, their nucleation, orientation, and dynamic remodelling, thus represents a critical regulatory point for cell wall architecture. Several aspects of microtubule organisation exhibit features potentially consistent with condensate-based mechanisms, and recent work in animal cells has established that key microtubule-associated proteins form functional condensates through LLPS.

In animal cells, microtubule nucleation occurs at centrosomes, which contain phase-separated pericentriolar material (PCM) that concentrates tubulin, γ -tubulin ring complexes (γ -TuRCs), and nucleation-promoting factors including Targeting Protein for Xklp2 (TPX2) and pericentrin (Woodruff *et al.*, 2017). PCM phase separation is driven by multivalent interactions among scaffold proteins containing coiled-coil and IDRs, creating a concentrated environment that dramatically enhances microtubule nucleation rates. The PCM exhibits characteristic condensate properties including spherical morphology driven by surface tension, rapid internal molecular exchange demonstrated by FRAP analysis showing sub-minute recovery times, fusion behaviour when centrosomes are brought into proximity, and dissolution upon hexanediol treatment (Woodruff *et al.*, 2017).

Beyond centrosomal PCM, microtubule plus-end tracking proteins (+TIPs) in animal cells form phase-separated condensates that organise microtubule dynamics. End-binding protein 1 (EB1), the core scaffold of the +TIP network, undergoes LLPS to constitute the microtubule plus-end machinery (Song *et al.*, 2023). EB1 condensation is driven by multivalent interactions among its calponin homology domain, IDR linker region, and homodimerisation domain, with charged lysine and arginine residues in the linker region being essential for phase separation. Phase-separated EB1 provides a compartment that enriches tubulin dimers and recruits other +TIPs including TIP150,

MCAK, and CLASPs through SxIP motif-mediated interactions. EB1 condensates exhibit characteristic of LLPS: concentration-dependent assembly, rapid FRAP recovery, coalescence behaviour, and sensitivity to hexanediol treatment. Crucially, LLPS-deficient EB1 mutants (EB1-KR6Q) fail to form condensates containing partner proteins and cannot support proper chromosome segregation during mitosis, demonstrating the functional significance of +TIP condensation (Song *et al.*, 2023; Yang *et al.*, 2024).

Another +TIP, the Cytoplasmic Linker-Associated Protein (CLASP), which promotes microtubule rescue and stabilisation, similarly form condensates that regulate microtubule organisation. In animal cells, CLASP2 accumulates in protein condensates at both the cell cortex and microtubule plus ends (Jia *et al.*, 2024). These CLASP2 condensates physically contact cortical scaffold condensates (formed by proteins including ELKS1 and LL5 β) through competitive binding interactions mediated by a conserved CLASP-binding motif. This condensate-condensate interaction determines cortical microtubule targeting, with phosphorylation of CLASP2 modulating the dynamics of these interactions to spatiotemporally navigate microtubule growth direction. The system represents a tuneable multiphase condensate network that coordinates microtubule organisation with cortical landmarks.

In contrast to the established condensate behaviour of their animal homologues, whether plant +TIPs including EB1 and CLASP form bona fide phase-separated assemblies remains untested. Plant EB1 and CLASP accumulate at growing microtubule ends where they regulate polymerisation dynamics and may mediate interactions with cortical landmarks (e.g., local tension; (Lian *et al.*, 2025; Nakamura *et al.*, 2018), exhibiting concentration-dependent clustering and rapid exchange with cytoplasmic pools (Hess and Joseph, 2025). These properties are consistent with but not yet validated as phase separation using established condensate criteria such as those mentioned (e.g., FRAP, hexanediol).

Plant cells lack centrosomes but instead nucleate cortical microtubules from γ -tubulin complexes distributed along existing microtubules and at the plasma membrane (King and Petry, 2020; Lindeboom et al., 2019; Murata et al., 2005; Walia et al., 2014). Whether these nucleation sites represent condensate-like assemblies analogous to animal PCM remains unknown. The augmenting microtubule nucleation (AUGMIN) complex, which recruits γ -TuRCs to microtubule surfaces for branching nucleation, forms discrete puncta along microtubules in plant cells (Alfaro-Aco et al., 2020; Liu et al., 2014; Tariq et al., 2020). AUGMIN puncta exhibit some features potentially consistent with condensate formation: concentration-dependent assembly, apparent dynamic exchange of components based on fluorescence imaging, and sensitivity to cellular conditions affecting molecular crowding. In reconstituted systems using *Xenopus* egg extracts, AUGMIN and the spindle assembly factor TPX2 form distinct puncta on microtubules where γ -TuRCs are recruited for branching microtubule nucleation (Alfaro-Aco et al., 2020; Liu et al., 2014; Tariq et al., 2020). However, whether AUGMIN puncta in plant or animal cells form through LLPS or represent stable protein complexes assembled through conventional binding mechanisms has not been rigorously determined using condensate criteria.

Molecular mechanisms connecting cytoskeletal dynamics to condensate behaviour in the broader cellular context remain incompletely understood. Several potential modes of interaction exist: direct binding of condensate components to cytoskeletal filaments, regulation of condensate material properties by mechanical forces transmitted through the cytoskeleton and coordinated regulation by common upstream signals including calcium and ROS. Processing bodies (P-bodies) and stress granules, well-characterised cytoplasmic condensates involved in mRNA turnover regulation and rescue, interact with cytoskeletal networks for proper localisation and dynamics. In animal cells, stress granule positioning depends on both actin retrograde flow and microtubule-mediated adhesive interactions (Böddeker et al., 2023). In Arabidopsis, P-body component DCP1 associates with the WAVE/SCAR complex and influences actin-dependent tissue growth (Liu et al., 2023b).

These observations suggest that mechanical signals originating from cell wall perturbations may influence cytoplasmic condensate behaviour through modulation of molecular crowding and cytoskeletal organisation, potentially providing a mechanism to coordinate cell wall biosynthesis by affecting stability of relevant mRNAs with broader cellular stress responses. Interestingly, as explained below, P-bodies are rich in RNAs involved in RNA regulation (see also **Supplemental File 3** for a detailed meta-analysis of datasets obtained in (Liu et al., 2024a)). However, direct connections between cell wall and regulation of specific intracellular condensates involved in wall dynamics have not been demonstrated and remain an important area for future investigation.

5. Condensates in cell wall integrity

Whilst the biosynthetic pathways described above operate during normal development, plants must continuously monitor cell wall integrity and respond to perturbations arising from mechanical stress, pathogen attack, osmotic challenge, and developmental transitions. Critically, CWI perturbations also trigger secondary consequences, notably osmotic stress and molecular crowding, that can independently promote condensate formation through biophysical mechanisms distinct from direct wall sensing. This section systematically examines how condensates participate in each tier of the CWI pathway: extracellular and membrane-proximal condensates involved in signal detection, membrane-associated and cytoplasmic condensates contributing to signal relay, and nuclear condensates mediating transcriptional and post-transcriptional responses. Throughout, we distinguish between condensates that directly sense wall properties and those responding indirectly to downstream consequences of CWI perturbations.

5.1 Extracellular and membrane condensates in CWI sensing

Elevated salt and temperature, two common stresses affecting CWI through osmotic effects, trigger an enhanced RALF–pectin phase separation described above, leading to promiscuous clustering and massive endocytosis of diverse plasma membrane proteins including not only FER and LLG1 but also brassinosteroid receptor BRASSINOSTEROID INSENSITIVE 1 (BRI1), auxin transporters PIN–FORMED 2 (PIN2) and AUXIN RESISTANT 1 (AUX1) (Liu *et al.*, 2024c). For all other CWI receptors, no experimental evidence currently exists for phase separation involvement, condensate recruitment, or condensate–mediated signalling.

Beyond true LLPS, the cell wall exhibits additional organisational mechanisms that achieve spatial precision through distinct biophysical principles. These mechanisms, whilst not constituting phase–separated condensates, contribute to CWI maintenance and merit brief consideration for context. Pectin polymers undergo methylesterification–dependent structural transitions that organise cell wall architecture. Super–resolution microscopy revealed that homogalacturonan, a major pectin component, assembles into discrete nanofilament structures in cell walls (Haas *et al.*, 2020). These HG nanofilaments exhibit methylesterification–dependent transitions between quaternary structures: highly methylesterified HG forms compact, hexagonally packed filaments, whilst demethylesterification triggers conversion to a rectangular lattice with approximately 1.4–fold radial expansion, providing a driving force for cell wall remodelling during growth. Importantly, this represents a polymer packing transition rather than LLPS, as it lacks the hallmarks of typical condensate behaviour including fusion–fission dynamics, rapid internal molecular exchange, and 1,6–hexanediol sensitivity.

Charge–dependent pectin patterns also enable spatial organisation of cell wall proteins through binding interactions distinct from LLPS. PME16 generates partially demethylesterified homogalacturonan domains that serve as anchoring platforms for PEROXIDASE 36 (PRX36),

positioning this enzyme precisely within specific cell wall microdomains to enable localised wall loosening during seed mucilage release (Francoz et al., 2019). Again, this pectin-mediated protein scaffolding differs mechanistically from LLPS but demonstrates how charge patterning, a common feature of proteins undergoing LLPS, creates spatially restricted functional ‘condensate’ zones within the wall matrix.

More directly relevant to CWI, is a case of peroxidase-mediated cross-linking contributes to wall reinforcement: PRX9/PRX40 oxidatively cross-link extensins to maintain anther and microspore CWI, with mutants displaying complete male sterility from compromised walls (Jacobowitz *et al.*, 2019), whilst PRX64 polymerises lignin at CASP-marked domains to form Casparian strip diffusion barriers in root endodermal cells (Lee et al., 2013). These peroxidase-scaffold systems represent complementary organisational mechanisms to phase separation, achieving spatial precision through enzymatic recruitment rather than typical phase transitions included in the conceptual framework of **Supplemental File 1**.

Whilst membrane-associated protein complexes mediating CWI have been extensively characterised in plants, whether condensates contribute to signal relay organisation remains an open question requiring systematic investigation. Membrane surfaces can influence condensate behaviour through several biophysical mechanisms demonstrated primarily in reconstituted systems and animal cell models. Lipid bilayers where CWI receptors reside can template condensate formation by increasing local protein concentration through two-dimensional confinement, reducing dimensionality from three-dimensional cytoplasm to two-dimensional membrane leaflets and thereby lowering the critical concentration threshold for phase separation. Banjade and Rosen (2014) demonstrated this principle using the nephrin-Nck-N-WASP system, where membrane attachment enabled condensate formation at approximately 50-fold lower protein concentrations than required for three-dimensional droplets in solution, with phase separation occurring in a two-

dimensional plane just above the membrane surface rather than representing binding of pre-formed three-dimensional droplets to membrane-attached receptors (Banjade and Rosen, 2014). Two-dimensional condensates exhibit distinct material properties compared to their three-dimensional counterparts, with membrane attachment restricting fusion dynamics and stabilising non-equilibrium shapes through capillary forces at the condensate-membrane-solution interface (Gouveia et al., 2022). Specific lipid-protein interactions provide additional mechanisms for recruiting condensate components to membranes, with phosphatidylinositol phosphate binding to pleckstrin homology domains recruiting proteins to specific membrane locations and potentially altering protein conformations to expose interaction sites that promote phase separation (Case et al., 2019).

In animal cell immune signalling, membrane-associated condensates organise signalling pathways through protein phase separation. In T cells, phosphorylated LAT (Linker for Activation of T cells) undergoes phase separation with cytoplasmic adaptor proteins Grb2 and Sos1 to form submicrometric-sized signalling clusters on the plasma membrane that concentrate kinases whilst excluding phosphatases, creating biochemically distinct zones that enhance signal transduction (Su et al., 2016). Whether similar membrane-condensate interactions contribute to plant CWI signal relay requires experimental investigation, particularly given unique plant cell constraints including rigid cell wall structure, distinct membrane lipid compositions enriched in phytosterols rather than cholesterol, and plant-specific signalling architectures.

Taken together, the diverse behaviours, spanning RALF-pectin LLPS to polymer structural transitions to protein scaffolding, collectively illustrate the cell wall as a highly organised, multimodal environment where different organisational principles cooperate to maintain integrity and enable dynamic remodelling during growth and stress responses. However, only RALF-pectin condensates currently satisfy the criteria for *bona fide* LLPS in the extracellular space. The capacity of polyanionic polysaccharides to induce protein phase separation through multivalent electrostatic interactions is

well established in animal systems: heparan sulphate drives LLPS of basic fibroblast growth factor on cell surfaces to facilitate receptor signalling (Xue et al., 2022), whilst heparin induces chemokine condensates that establish concentration gradients for directional cell migration (Yu et al., 2024). The RALF–pectin condensates demonstrate that analogous mechanisms operate in the plant apoplast (Liu *et al.*, 2024c). Whether pectin–mediated protein recruitment in other contexts or during wall loosening, softening, or fluidisation of the wall, as exemplified by the PME16–PRX36 system, represents a continuum with LLPS or a mechanistically distinct scaffolding phenomenon remains to be determined through systematic (biophysical) characterisation. Future research should systematically investigate how plant membrane lipid compositions affect potential condensate–lipid coupling affecting delivery of cell wall material, and whether condensate formation provides functional advantages for CWI signal relay beyond conventional protein complex assembly.

5.2 Cytoskeletal condensates in CWI

When CWI is compromised, cells must rapidly reorganise their cytoskeleton to redirect cellulose deposition, reinforce wall architecture, and activate compensatory responses. Emerging evidence suggests that condensates may participate in coordinating these stress–induced cytoskeletal reorganisations, although the mechanisms remain incompletely understood. The clearest example of condensate–mediated cytoskeletal regulation during CWI comes from poplar, where the Ca^{2+} –binding protein PagPCaP1a undergoes LLPS in response to salt stress (Lian *et al.*, 2025). Salt stress creates osmotic imbalances that alter turgor pressure and cell wall mechanics, triggering adaptive cellular responses including cytoskeletal reorganisation. Under these conditions, PagPCaP1a forms liquid–like condensates that regulate microtubule depolymerisation, providing a mechanism for rapid cytoskeletal remodelling in response to stress conditions affecting CWI. This represents direct experimental evidence that phase separation can modulate cytoskeletal dynamics relevant to CWI maintenance. The Ca^{2+} –dependent nature of PagPCaP1a condensation suggests a potential mechanism for integrating Ca^{2+} signalling, a central component of CWI stress responses

downstream of mechanosensitive Ca^{2+} channels, with cytoskeletal reorganisation through condensate-mediated regulation.

The connection between cytoskeletal dynamics and other stress-responsive condensates provides additional potential mechanisms for CWI coordination. The actin cytoskeleton itself reorganises rapidly in response to cell wall perturbations and mechanical stress (Henty-Ridilla et al., 2013; Pleskot et al., 2013), and this reorganisation may affect P-body localisation and dynamics. This raises the possibility that mechanical signals originating from cell wall perturbations propagate through the cytoskeleton to influence condensate behaviour, potentially through alterations in molecular crowding, changes in cytoskeletal tension that affect condensate material properties, or sequestration of condensate components by cytoskeletal structures. However, the directionality of this relationship—whether cytoskeletal changes drive condensate reorganisation, condensates direct cytoskeletal responses, or bidirectional regulation occurs—remains unclear and likely varies depending on the specific stress and cellular context.

Current evidence suggests that cytoskeletal condensates participate in CWI stress responses primarily through the documented example of PagPCaP1a-mediated microtubule regulation during salt stress, with other proposed mechanisms including CSI1–CESA coupling and MAP/+TIP organisation remaining speculative. Several critical questions require experimental investigation: Do CSI1 puncta represent bona fide phase-separated condensates, and if so, do they respond dynamically to CWI perturbations? Do +TIP proteins (or MAPs in general) undergo stress-induced condensation to coordinate rapid cytoskeletal reorganisation? How do cytoskeletal dynamics influence the behaviour of other stress-responsive condensates including P-bodies and stress granules during CWI? Addressing these questions will require systematic application of condensate validation criteria combined with functional analyses demonstrating that condensate formation is necessary for proper CWI responses.

5.3 Condensates organising transcriptional and translational responses to CWI perturbations

CWI perturbations trigger both direct responses that specifically monitor cell wall status and indirect responses to secondary consequences including osmotic stress and molecular crowding. Biomolecular condensates participate in both response types, though with varying degrees of specificity. Indirect responses could occur when CWI perturbations cause osmotic stress, alter turgor pressure, or induce molecular crowding, which then trigger condensate formation through general stress mechanisms. The molecular basis could involve changes in cellular biophysics: osmotic stress from external salt exposure or internal turgor changes caused by cell wall weakening increases effective macromolecular concentration through water loss and reduced cellular volume (Delarue et al., 2018). This molecular crowding reduces the critical concentration threshold for phase separation (see also **Supplemental File 1**, Csat parameter) through excluded volume effects that decrease available space for each molecule, effectively increasing local concentration and reducing the entropic favourability of the homogeneous mixed state, thereby promoting condensate formation (André and Spruijt, 2020; Choi et al., 2020; Das et al., 2015).

P–bodies, to execute their functions in RNA expression and turnover regulation, contain core decapping machinery including DCP1, DCP2, and DCP5, along with deadenylases, exonucleases, and translational repressors (Xu and Chua, 2009; 2011). These condensates serve as hubs for mRNA decay and translational repression during stress conditions (Liu et al., 2024a). DCP5 plays a particularly important role during osmotic stress, forming specialized granules independent of P–bodies called DCP5 osmotic stress granules (DOSGs) through the above–mentioned molecular crowding–induced phase separation (Wang et al., 2024b). Furthermore, as mentioned above, recent proximity–RNA capture approaches revealed that P–bodies preferentially recruit transcripts encoding hub genes in regulatory networks, including cell wall regulation (Liu et al., 2024a). One

probable scenario is that when DOSGs are formed through sequestration of DCP5, P–bodies are destabilised and release mRNAs of which some are cell wall–related. These RNAs could then be translated to enhance the CWI response, independent from nuclear transcriptional programs. In particular, P–bodies contain also cell wall–modifying enzymes, such as PMEIs (Liu *et al.*, 2024a) and further enriched in genes regulating pectin (**Supplemental File 3**). This evidence point to a possible role of P–bodies or other condensates like DOSGs into the regulation of CWI through a combination of RNA turnover regulation and osmotic effects. P–bodies also interact extensively with the actin cytoskeleton through DCP1 associations with the WAVE/SCAR complex, with disruption of actin dynamics affecting P–body formation and localization (Liu *et al.*, 2023c). This cytoskeletal connection may be functionally significant for CWI, as actin organisation itself responds to cell wall perturbations and mechanical stress (Bashline *et al.*, 2014), though whether P–bodies could regulate mRNA availability for CWI–related proteins as part of specific wall status monitoring versus general stress responses requires systematic investigation.

Under severe osmotic stress, subgroup I SUCROSE NON–FERMENTING 1–RELATED PROTEIN KINASE 2 (SnRK2) kinases form cytoplasmic condensates through their C–terminal IDRs, responding to molecular crowding caused by cell volume reduction (Yuan and Zhao, 2025). These kinase condensates spatially segregate SnRK2s from their negative regulator, the PP2C phosphatase ABI1, releasing SnRK2s from inhibition and allowing activation through autophosphorylation. While osmotic stress frequently accompanies CWI perturbations during conditions that weaken turgor pressure, SnRK2 condensates respond to molecular crowding rather than directly monitoring CWI, representing an indirect connection to CWI maintenance. TSN–containing stress granules are essential for activating SnRK1 kinase (Gutierrez-Beltran *et al.*, 2021), a central regulator of energy homeostasis that impacts CWI–related processes through effects on carbohydrate metabolism and cell wall precursor availability. These two evidence illustrate how condensate–mediated signalling can indirectly link cellular energy status to cell wall biosynthesis.

A plausible, but explicitly hypothetical, model is that CWI receptors such as FER or THE1, triggering kinase cascades involving BIK1, Ca^{2+} –dependent protein kinases, or RAPIDLY ACCELERATED FIBROSARCOMA (RAF)–like kinases that modify condensate components. For example, under osmotic stress, subclass I SnRK2 kinases localise to P–bodies and phosphorylate the mRNA decapping activator VARICOSE (VCS), modulating mRNA decay (Soma et al., 2020). SERRATE, a component of the microprocessor complex, is also phosphorylated by SnRK2 kinases under stress conditions (Fàbregas et al., 2020). Interestingly, these targets are phosphorylated by SnRK2 kinases both downstream of RAF–like kinases and in response to osmotic stress (Kuhn et al., 2024; Soma et al., 2020). Precedents from osmotic and stress signalling suggest that distinct pathways may converge on overlapping condensate regulators, but whether similar convergence occurs downstream of cell wall sensors remains unknown.

More direct evidence for connections between condensates and CWI involves the RNA–binding protein GLYCINE–RICH RNA–BINDING PROTEIN 7 (GRP7), which is directly phosphorylated by FER (Wang et al., 2020; Xu et al., 2024a). GRP7 forms stress granules under temperature stress, where FER–mediated phosphorylation enhances its LLPS and modulates translation (Xu et al., 2024a). Noteworthy, GRP7 is a known component of P–bodies and stress granules (Liu et al., 2024a). However, whether FER–GRP7 phosphorylation also regulates P–body function, and if so whether this links to direct CWI surveillance versus broader stress responses, remains experimentally undemonstrated.

Nuclear condensates formed by transcription factors represent stress–responsive compartments that indirectly affect CWI through osmotic stress pathways. SEUSS, a transcriptional co–repressor, rapidly forms liquid–like nuclear condensates under hyperosmotic conditions (Wang et al., 2022). SEUSS condensation depends on its first IDR1, which acts as a crowding sensor: when cells experience osmotic stress, molecular crowding triggers IDR1 conformational changes that expose

interaction motifs promoting multivalent interactions and phase separation. SEUSS condensation is indispensable for osmotic stress tolerance, and loss of SEUSS dramatically compromises expression of stress-responsive genes. Whilst some SEUSS target genes encode proteins affecting cell wall–membrane interactions, SEUSS condensates respond to molecular crowding rather than directly sensing cell wall properties.

DREB2A (Dehydration–Responsive Element Binding protein 2A) belongs to the AP2/ERF transcription factor family and activates expression of stress-responsive genes including cell wall–modifying factors such as expansins, pectin methylesterases, and xyloglucan endotransglucosylases/hydrolases (Bi et al., 2024; Mizoi et al., 2019; Sakuma et al., 2002). While DREB2A regulates cell wall–modifying enzymes, it responds to dehydration and heat stress rather than directly monitoring wall status, representing an indirect relationship to CWI maintenance. Recent work though in rice identified that the cold stress–sensing protein OsSRO1c forms nuclear condensates via its IDR and recruits the transcription factor OsDREB2B into these condensates to enhance its transcriptional activity (Hu et al., 2024), illustrating how transcription factors can be recruited into condensates formed by other regulatory proteins. Whether similar mechanisms operate for DREB2A in Arabidopsis during heat or drought stress remains to be investigated.

Developmental transcription factors regulating secondary cell wall formation during vessel and fibre differentiation, including NAC domain proteins (SND1, NST family, VND proteins) and their downstream MYB targets (MYB46, MYB83) coordinate expression of cellulose synthases, xylan biosynthetic enzymes, and lignin biosynthetic genes (Nakano *et al.*, 2015; Zhong *et al.*, 2007b). While these factors contain IDRs that could potentially support phase separation, no experimental evidence currently demonstrates their condensate behaviours.

Beyond individual transcription factors, larger nuclear bodies may organise gene regulatory programmes. The most speculative possibility involves transcriptional condensates at enhancers

and super-enhancers, a phenomenon extensively studied in animal systems but poorly characterised in plants. In mammals, transcriptional co-activators such as Mediator complex subunits and BRD4 form phase-separated condensates at super-enhancers that drive high-level expression of cell identity genes (Sabari et al., 2018). These condensates concentrate transcription factors, co-activators, and RNA polymerase II, creating 'transcription factories' that enable robust, coordinated expression of target genes. Whether analogous mechanisms operate in plants for coordinating cell wall biosynthetic gene expression remains unknown.

In *Arabidopsis* and other plant species, genes encoding CESA and cellulose synthase-like proteins (Csl) are dispersed across multiple chromosomes rather than forming genomic clusters (Holland et al., 2000; Pancaldi et al., 2022). Phylogenomic analyses across 242 plant genomes reveals that Cesa genes predominantly exist as singletons or small tandem arrays, with closely related paralogs typically mapping to different chromosomal locations (Pancaldi et al., 2022). Despite this physical separation, these genes exhibit remarkably coordinated expression patterns during cell wall biogenesis, with primary wall Cesa genes (Cesa1, Cesa3, Cesa6-like) and secondary wall Cesa genes (Cesa4, Cesa7, Cesa8) forming distinct co-expression modules (Brown et al., 2005; Mutwil et al., 2009; Persson et al., 2005). This transcriptional coordination is mediated through shared cis-regulatory elements, including binding sites for MYB, WRKY, and NAC family transcription factors (Heidari et al., 2019; Vandepoele et al., 2009), as well as through master regulators such as the NAC-MYB transcriptional network governing secondary cell wall biosynthesis (Nakano et al., 2015)(Nakano et al., 2015). Whether transcriptional condensates contribute to this coordinate regulation, and if so, through what mechanism given the dispersed genomic locations, remains an open question requiring systematic investigation.

6. How to bridge knowledge gaps in the field: an AI-based proposal

Accumulating evidence suggests that condensates operate at multiple levels of cell wall biology. At the same time, this emerging framework exposes substantial knowledge gaps. Understanding condensate involvement in cell wall biosynthesis and CWI poses challenges distinct from those encountered in general condensate biology. Plant cells must integrate condensate dynamics with cell wall mechanics and turgor-driven forces and associated osmotic-pressure effects through mechanisms that remain poorly characterised. One theoretical possibility is that condensates at membrane interfaces generate elastocapillary forces that help maintain plasma membrane–cell wall adhesion during mechanical stress, alongside other non-mutually exclusive mechanisms such as cytoskeletal anchoring or membrane–wall tethering complexes. However, experimental support for these ideas in plant systems is currently lacking. Moreover, CWI perturbations can induce condensate formation across multiple compartments, including extracellular RALF–pectin assemblies, P-bodies and stress granules, and nuclear condensates, raising questions about cross-compartment coordination.

Computational methods offer conceptual frameworks for addressing these gaps, though their application in this field remains limited. Network inference from multi-omics datasets could, in principle, identify regulatory relationships between cell wall sensors and condensate-associated proteins, complementing labour-intensive genetic screens. Machine learning approaches integrating sequence features with structural predictions may help prioritise candidate condensate components (e.g., IDR-rich) for experimental testing, whilst deep learning-based image analysis could quantify condensate dynamics (e.g., reversibility) across compartments under defined CWI perturbations. However, these approaches face recurring limitations: plant-specific training datasets

are scarce, most algorithms are optimised for mammalian systems, and predictions require extensive experimental validation under physiologically relevant conditions. Below, we describe a potential roadmap.

6.1 Predicting condensation-prone components of CWI pathways

Current phase separation predictors show variable performance for plant proteins, largely because their training datasets derive predominantly from human disease-associated proteins. PSAP, for example, uses random forest classifiers trained on human proteome features and achieves a receiver operating characteristic (ROC) area under the curve (AUC) of 0.88 in cross-validation (van Mierlo et al., 2021). Whilst this score indicates good classification performance, it does not guarantee biological relevance when applied to plant proteins with distinct evolutionary and biophysical constraints. Furthermore, although predictors can identify IDRs or multivalent motifs in WAKs and mechanosensitive channels (**Supplemental File 4**), these predictions have rarely been validated through phase separation assays under normal or CWI-relevant conditions such as mechanical stress, osmotic perturbation, or altered ion composition. Critically, current tools generally classify proteins as phase separation-prone or not, without distinguishing scaffold proteins that drive condensate assembly from client proteins that partition into pre-existing condensates. FER exemplifies this limitation: despite containing disordered regions, it does not nucleate condensates but instead partitions into RALF-pectin assemblies (Liu et al., 2024c).

Transformer-based approaches such as Seq2Phase represent advances in distinguishing scaffold and client behaviour (Miyata and Iwasaki, 2024), whilst structure-informed predictors such as catGRANULE 2.0 enable single amino acid resolution predictions and mutation effect assessment (Monti et al., 2025). However, these models were developed primarily using human datasets, and their performance on plant proteins, remains insufficiently validated. More broadly, none of the current predictors account for the biophysical parameters central to cell wall biology, including

mechanical load, turgor pressure, or dynamic ion fluxes. Hence, systematic experimental pipelines are needed to test computational predictions or even build new ones. Until such frameworks exist, prediction tools should be viewed as prioritisation instruments rather than sources of biological inference. Notably, despite the central role of RNA in many condensates, no experimentally validated tool exists for predicting condensation-prone RNAs.

6.2 Omics-based inference of CWI-condensate networks

Network inference from transcriptomic or proteomic data faces intrinsic limitations when applied to condensate biology. Condensate reversibility and dynamics occur on timescales of seconds to minutes, whereas transcriptomic changes unfold over hours. As a result, expression-based networks can only indirectly reflect processes that regulate phase separation. Whilst graph neural networks developed for spatial transcriptomics offer useful frameworks (Dong and Zhang, 2022; Long et al., 2023), plant datasets remain limited, particularly under controlled mechanical or wall-damaging conditions. Integrating transcriptomics with proteomics and phosphoproteomics could help identify signalling cascades linking CWI sensors to condensate regulation. Distinguishing proteins that directly sense wall status from those responding indirectly to osmotic stress or molecular crowding remains a central challenge. Network inference can reveal correlations affecting protein concentration thresholds for phase separation, but it cannot establish causality or functional relevance without targeted genetic perturbations and temporal resolution.

6.3 AI-based phenotyping of CWI-condensate interactions

AI-driven phenotyping of condensates in plant systems remains underdeveloped. Cell wall autofluorescence, tissue thickness, and architectural complexity complicate fluorescence imaging and automated detection. Whilst deep learning models achieve high accuracy for condensate detection in mammalian cells, comparable validation in plants has not yet been reported. Existing tools such as PlantSeg were developed for cell segmentation rather than subcellular condensate

detection and have not been applied to the contexts explored herein (Wolny et al., 2020). At larger scales, atomic force microscopy (AFM) enables quantitative mapping of cell wall mechanical properties, but direct integration of mechanical maps with condensate localisation or dynamics has not been demonstrated. Correlating AFM-derived wall mechanics with condensate behaviour could provide insight into whether condensates preferentially form at mechanically defined domains (Peaucelle *et al.*, 2011), but such multimodal analyses remain unexplored. Field-scale phenotyping represents a longer-term challenge. Linking condensate dynamics to environmental sensing or crop performance would require validated condensate biomarkers of cell wall status and imaging platforms compatible with intact tissues, resources that are not yet available.

6.4 Toward multi-scale integration of CWI and condensate biology

Multi-modal AI frameworks that integrate sequence features, structural predictions, molecular networks, and phenotypic data offer a conceptual path forward, but practical application to plant systems is constrained by data scarcity. Deep learning architectures have achieved transformative success in protein structure prediction with AlphaFold2's Evoformer architecture (Jumper et al., 2021) and protein language modelling with transformer-based approaches such as ProtTrans (Elnaggar et al., 2022). Graph-based integration frameworks such as MOGONET (Multi-Omics Graph cOnvolutional NETworks) have been developed for combining multi-omics data, including gene expression, DNA methylation, and microRNA profiles, to understand complex traits in human disease contexts (Wang et al., 2021b). Unlike human research, plant condensate biology lacks comprehensive datasets linking genetic variation, condensate behaviour, and physiological outcomes. Moreover, predicting condensate behaviour requires capturing high-dimensional parameters including post-translational modifications, material state transitions, assembly kinetics, and responses to combined mechanical and osmotic stresses. As a result, current AI methods cannot yet predict condensate function under cell wall-relevant conditions. Substantial experimental investment will be required to generate training datasets spanning developmental stages, genetic

backgrounds, and environmental perturbations before these approaches can move beyond hypothesis generation.

7. Concluding remarks

Across scales, a unifying principle emerges: condensates may function as dynamic mechanochemical integrators that couple cell wall status to intracellular signalling through rapid, reversible reorganisation of proteins and RNAs. AI-based approaches provide powerful tools for exploring this possibility, but only when grounded in plant-specific, mechanically perturbed, and temporally resolved experimental data. Establishing such foundations will be essential to further our understanding of both CWI and condensates' biology.

In this review, we traced cell wall biology from fundamental biosynthetic mechanisms through stress-responsive integrity maintenance pathways, demonstrating how biomolecular condensates emerge as organisational principles operating at multiple regulatory levels. Current evidence for condensate roles in cell wall dynamics reveals a hierarchy of mechanistic certainty. For example, RALF-pectin condensates represent the strongest example of direct CWI-condensate coupling (Liu *et al.*, 2024c). Beyond this validated example, proposed condensate roles in Golgi-localised biosynthetic machinery organisation, TGN cargo sorting, cytoskeletal nucleation complexes, and transcriptional regulation require further systematic experimental investigation.

A critical distinction emerges between condensates that directly monitor cell wall status and those responding to downstream consequences of perturbations (**Table 1**). Nuclear condensates formed by SEUSS respond to molecular crowding during osmotic stress, facilitating transcriptional activation of stress-tolerance genes through IDR-driven phase separation (Wang *et al.*, 2024a). Similarly, in rice, OsSRO1c forms nuclear condensates that recruit the transcription factor OsDREB2B under cold stress, enhancing transcriptional activity of cold-responsive genes including *COLD1* (Hu *et al.*,

2024), though this represents a temperature–sensing mechanism rather than a direct CWI response. Cytoplasmic stress granules and P–bodies that sequester translationally stalled mRNAs respond to molecular crowding and osmotic stress rather than directly sensing wall mechanical or compositional properties. Yet, P–bodies selectively sequester mRNAs and proteins encoding cell wall–modifying enzymes (**Supplemental File 3**; meta–analysis of (Liu *et al.*, 2024a)). The distinction between direct CWI sensing and osmotic stress or cell volume changes matters for understanding regulatory specificity: How do cells distinguish mechanical wall damage from osmotic stress or pathogen attack when multiple stress types trigger overlapping condensate responses? What molecular signatures enable condensates to activate selectively appropriate transcriptional programmes for specific perturbations? Do feedback mechanisms exist that coordinate condensate dynamics across cellular compartments, for example, from extracellular RALF–pectin assemblies through cytoplasmic signalling condensates to nuclear transcriptional hubs?

Condensate–based biotechnology for engineering enhanced cell wall properties faces substantial current limitations despite theoretical promise. Whilst some small molecules selectively partition into specific condensates based on their physicochemical properties, particularly features related to solubility and hydrophobicity (Ambadi Thody *et al.*, 2024; Kilgore and Young, 2022), recent work demonstrates that partitioning is correlated among disparate condensates, indicating underlying physical similarities rather than condensate–specific targeting (Ambadi Thody *et al.*, 2024). The potential for engineering climate–adapted crops through condensate manipulation—for example, by modulating RALF–pectin condensate formation to enhance CWI sensing or by altering transcription factor condensate dynamics to improve stress tolerance—remains speculative without demonstration of improved agronomic performance under variable field conditions. Divergent disordered sequences driving phase separation may provide mechanisms for natural selection to tailor condensate material properties to specific environmental niches, but rational design of

condensates with desired properties requires far deeper understanding of sequence–function relationships than currently available.

AI integration with condensate biology shows promise for accelerating discovery but faces significant validation challenges. Machine learning algorithms report high accuracy for identifying phase–separating proteins; however, performance metrics typically derive from non-plant datasets, with systematic plant validation remaining limited. Network inference algorithms generate hypotheses about regulatory connections between CWI sensors and condensate–forming proteins but cannot establish causality between computational correlations and biological mechanisms without experimental validation. High–throughput imaging approaches could quantify condensate dynamics across multiple cellular compartments during CWI perturbations, but training datasets for plant condensate detection in complex tissue environments do not yet exist. Multi–modal integration frameworks offer theoretical potential for synthesising diverse data types, but practical deployment requires substantial investment in generating comprehensive plant–specific training datasets spanning genetic backgrounds, developmental stages, and environmental conditions.

Future progress requires interdisciplinary approaches integrating cell biology, biophysics, computational modelling, and field phenotyping. Key experimental priorities include: (1) systematic characterisation of condensate formation by CWI pathway components under relevant stress conditions; (2) temporal profiling capturing both rapid post–translational regulation of condensates and slower transcriptional responses; (3) genetic dissection of condensate contributions to CWI through targeted perturbations of phase separation behaviour; and (4) demonstration that condensate modulation improves stress tolerance in crop plants under field conditions. Computational priorities include generating plant–specific training datasets for phase separation prediction algorithms, validating network inference approaches through experimental perturbation studies, and developing imaging analysis pipelines adapted to plant tissue architecture and autofluorescence. Success will require moving beyond cataloguing condensate systems towards

understanding how condensate dynamics integrate with other regulatory mechanisms—protein complex assembly, membrane domain organisation, transcriptional networks—to achieve robust and adaptive cell wall dynamics maintenance across diverse environmental conditions.

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Competing interests

The authors declare no conflict of interests.

Declaration of generative AI and AI–assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) in order to assist with comprehensive literature search, reference formatting, identification of over-claiming during manuscript preparation, generation and refinement of scientific language. After using this

tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Author contributions

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Table 1. Condensates linked to CWI. Summary of biomolecular condensates with direct and indirect connections to CWI maintenance. Direct connections involve condensates that form at cell wall interfaces, directly sense wall properties, or participate in immediate CWI responses. Indirect connections affect CWI through osmotic stress, molecular crowding, or transcriptional regulation of CWI-related genes. All entries represent experimentally validated condensates identified through live-cell imaging, biochemical analysis, or genetic studies. CWI = cell wall integrity; DOSGs = DCP5 osmotic stress granules.